Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/29/2026
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Project Title:	Creating Non-Human Tended Microphysiological Systems (PI: Vunjak-Novakovic)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH Directed Studies and Initiatives
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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	Microphysiological systems (MPS), also known as organs-on-chips, are engineered to recapitulate organ-level and systemic functions in the context of human patho/physiology. Recent studies are increasingly demonstrating the potential of MPS to model the impact of space stressors (i.e., radiation, microgravity, oxidative stress) on astronauts' health, towards identifying the health risks and developing effective countermeasures. The current state of the art in the design and utilization of MPS for space research involves two major unmet needs: (i) rigorous validation of their physiological relevance, and (ii) compatibility with manned and unmanned flight missions. We propose to address these two needs by engineering the next-generation human MPS platform that will combine high biological fidelity of experimental conditions (emulating human physiology on Earth and in space) with configurable design and automated operation (for broad and high impact on addressing the effects of space stressors). To this end, we will focus on		

Task Description:	recapitulating human physiology in studies of tissue systems that are of highest interest for spaceflight; in situ sensing/monitoring, and automated operation. We will focus on four organ systems that are of the highest interest for space research because of their documented adverse effects on astronauts: bone marrow (BM); cardiac muscle (CM); small artery; and blood-retinal barrier (BRB); individually, and in combination. To enable individualized ("astronaut on chip") physiological studies, all tissues will be derived from the same human iPSCs and matured to recapitulate organ-level functions. To enable space studies, with and without crew support, we will design the platform for compatibility with the automated operation, with in situ sensing, data acquisition, and analysis. To validate the platform, we will demonstrate its capability to recapitulate human physiology on Earth. To study the physiological responses to spaceflight-related stressors, we will engineer all tissues from iPSCs gene-edited to systemically express oxidative stress, a known key stressor in space, and compare to unedited control tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our previous work on single- and multi-organ MPS platforms has a track record of innovative designs and the organ-level and systemic studies of human physiology. Uniquely, we engineered patient-specific tissues that are matured to adult-like phenotypes and maintained for weeks and months while allowing their communication via vascular circulation. This capability was established by maintaining the individual tissue/organ compartments while linking them by vascular flow to allow cross-talk, as in our body, and will now be advanced by the following contributions: (1) Gene-edited iPSCs expressing high oxidative stress, a widely recognized space stressor; (2) Incorporation of vascular flow, endothelial barriers, and circulating immune and hematopoietic cells generated by iPSC-derived bone marrow; (3) Implementation of automated techniques for tissue culture, in situ sensing and analysis; (4) Systematic validation of engineered tissues, individually and in the integrated platform, against their native counterparts; (5) Platform design enabling the transition to non-human-tended flight hardware. Our studies will specifically address the key risks identified by the NASA Human Research Program (HRP) (CV-202; CV-101; CV-102; Degen-1, 2, 3; IM-101; Pharm-601) in an "astronaut-on-a-chip" setting. We will genetically induce oxidative stress into iPSCs, interrogate the resulting changes in multiple tissues, identify targeted pathways and test protective countermeasures (CV-202). Furthermore, we will design the platform to enable an automated operation and transition to the flight hardware to enable the study of tissue responses to space stressors during crewed and uncrewed missions (CV-101; CV-102; Degen-1, 2, 3; IM-101). Finally, non-human tended MPS represent an open, standardized, and configurable platform for developing countermeasures to the effects of space environment (e.g., pharmacokinetics, pharmacodynamics, drug testing) (Pharm-601). As we are recapitulating the four organ systems of the highest interest to space research (bone marrow, cardiac muscle, small artery, and inner blood-retinal barrier), due to their adverse effects, our novel MPS platform will allow us to assess health risks during space travel and elucidate physiological and pathological mechanisms underlying spaceflight. This will ultimately advance precision medicine and accelerate the development of protective countermeasures.
Task Progress:	The main focus of our research in year 2 of the project was on the development and validation of microphysiological systems (MPS) comprised of single or multiple tissues with vascular barriers and flow connections, representing organ systems associated with spaceflight-related health risks. To this end, we have developed and tested a configurable plug & play platform with vascular lining and recirculating flow. The platform is equipped to support the bioengineered human organs/tissues of high interest for supporting human mission to space: inner blood–retinal barrier (iBRB), artery-on-chip (ARTOC), cardiac muscle, bone marrow, and vascular endothelium. For biological specificity, all tissues and vasculature were derived from iPS cells, enabling well-defined individualized studies. We report characterization of all tissues cultured individually or linked to each other at the molecular, cellular and functional levels (Aim 1 studies). To investigate the effects of spaceflight on human organs, we have gene-edited our parent iPSC line to induce oxidative stress via a knock-out (KO) of antioxidant gene Nrf2. Both the wild type a KO lines were differentiated into multiple cell lineages needed to engineer the four tissues systems and the vascular linings, and their multipotency and differentiation potentials were systematically compared. In addition, we conducted an initial set of experiments to measure the effects of acute exposure of iPSC-derived cells to radiation. These studies will be continued in year 3 into comprehensive and precise evaluations of the effects of this red risk on all four organ systems, individually and linked to each other (Aim 2 studies). Starting from the early inputs that we obtained from Dr Lucy Low, Chief Scientist at Axiom Space and Dr Stefanie Countryman, Director of BioServe Space, we continued to incorporate the design features that will facilitate the eventual transition into the flight hardware. This engineering analysis is directed to adoption of the 96 well plate footprint, conditioning of culture medium in a perfusion loop (instead of using the ground-based liquid-air interfaces), incorporating pumps and controllers into the platform (to avoid the tubing connections), towards a robust and precisely controlled platform with multiplexed readouts and automated operation suitable for uncrewed space studies. The work has been conducted in a highly collaborative manner in the two laboratories, and with cross-validation of the experimental components and methods between the labs. The responsibilities were divided between the teams at Columbia University and Duke University as follows: • Columbia University: development of the MPS platform, studies of engineered cardiac and bone marrow, derivation of Nrf2 KO cell line, radiation studies, transition to automated operation • Duke University: testing of the MPS platform, incorporation and studies of engineered inner blood–retinal barrier and artery on chip, studies of Nrf2 KO cells, and radiation studies. ORIGINAL PROJECT AIMS Aim 1: Validate the platform utility for noninvasive studies of human tissues Aim 2: Demonstrate capability to recapitulate responses of human tissues/organs to space flight stressors Aim 3: Develop a configurable MPS with in situ sensing and capability to be translated into flight hardware with automated operation. In year 2 of the project, we continued to make progress toward each of these aims and have accomplished the planned milestones and deliverables, as detailed in this report.

Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	He S, Zhu Y, Tavakol DN, Ye H, Lao YH, Zhu Z, Xu C, Chauhan S, Garty G, Tomer R, Vunjak-Novakovic G, Zou J, Azizi E, Leong KW. "Squidiff: Predicting cellular development and responses to perturbations using a diffusion model." Nat Methods. 2026 Jan;23(1):65-77. https://doi.org/10.1038/s41592-025-02877-y ; PMID: 41184550; PMCID: PMC12872408 , Jan-2026
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Awards	Vunjak-Novakovic G. "Research.com Top 1000 Female Scientists in the World based on h index: ranked 103rd in the US and 169th in the world, November 2025." Nov-2025
Awards	Vunjak-Novakovic G. "Web of Science, Highly cited scholar (top 1%), November 2025." Nov-2025